

LABORATÓRIO 7

Gold de risco por segmento

```
CREATE TABLE gold_fraud_risk
STORED AS PARQUET AS
SELECT
  segment,
  COUNT(*) AS total_transacoes,
  SUM(amount) AS valor_total,
  ROUND(AVG(amount), 2) AS ticket_medio,
  SUM(CASE WHEN is_fraud THEN 1 ELSE 0 END) AS qtd_fraudes,
  ROUND(100.0 * SUM(CASE WHEN is_fraud THEN 1 ELSE 0 END) / COUNT(*), 2) AS taxa_fraude_pct,
  SUM(CASE WHEN is_fraud THEN amount ELSE 0 END) AS valor_em_risco
FROM silver_transactions
GROUP BY segment;

SELECT * FROM gold_fraud_risk ORDER BY taxa_fraude_pct DESC;
```

segment	total	valor_total	ticket	fraudes	taxa%	valor_em_risco
High-Risk	9155	1664640.2447	181.83	705	7.70	126451.0691
Standard	29689	5479399.8467	184.56	655	2.21	106828.4191
Premium	61156	11235041.5762	183.71	473	0.77	81308.4380

High-Risk teve a maior taxa e quantidade de fraudes, enquanto Premium teve a menor taxa de fraude.

```
CREATE TABLE gold_daily_metrics
STORED AS PARQUET AS
SELECT year, month, day,
  COUNT(*) AS transacoes,
  SUM(CASE WHEN is_fraud THEN 1 ELSE 0 END) AS fraudes
FROM silver_transactions
GROUP BY year, month, day;

SELECT year, month, day, fraudes
FROM gold_daily_metrics
ORDER BY fraudes DESC
LIMIT 5;
```

2023	4	27	16
2023	12	22	13
2023	5	21	13
2024	5	23	13
2024	5	25	12

O objetivo dessa query é observar a distribuição diária das fraudes. O maior valor encontrado foi de 16 fraudes em 27/04/2023.